

HT-ProMon Series Installation Instructions



HT-ProMon-EMR
(Electromechanical Relay)



HT-ProMon-SSR
(Solid State Relay)

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1. Safety Precautions

IMPORTANT — READ BEFORE INSTALLATION OR OPERATION

This document contains critical safety information for the installation and operation of the HT - ProMon Series Heat Trace Controllers. Failure to follow these instructions may result in equipment damage, serious injury, or death.

1.1 Electrical Hazard

DANGER: Hazardous voltage is present inside the enclosure.

- Disconnect all power sources before opening, drilling, or servicing the unit.
 - Verify circuits are de -energized using a properly rated test instrument.
 - Ensure proper grounding and bonding of the enclosure and all connected equipment.
 - Installation must be performed in accordance with NEC (NFPA 70) and all applicable local codes.
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1.2 Installation Safety

- Installation and servicing must be performed by qualified personnel familiar with industrial electrical systems and heat tracing applications.
 - Provide a readily accessible disconnecting means within sight of the controller.
 - Use appropriately rated branch circuit protection for the connected load.
 - Do not install in environments exceeding the product's rated temperature or enclosure classification.
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1.3 Drilling & Conduit Entry Hazard

WARNING: Improper conduit entry can damage internal electronics and create unsafe

conditions.

- Follow all conduit entry restrictions outlined in this document (see Installation Notes).
- Protect internal components from debris during drilling or punching operations.
- Failure to comply may result in electrical shorts, equipment failure, or unsafe operation.

1.4 Thermal Hazard

- Heat trace systems may operate at elevated temperatures.
- Avoid contact with energized heating elements and associated piping.
- Allow sufficient cooling time before performing maintenance.

1.5 Maintenance & Handling

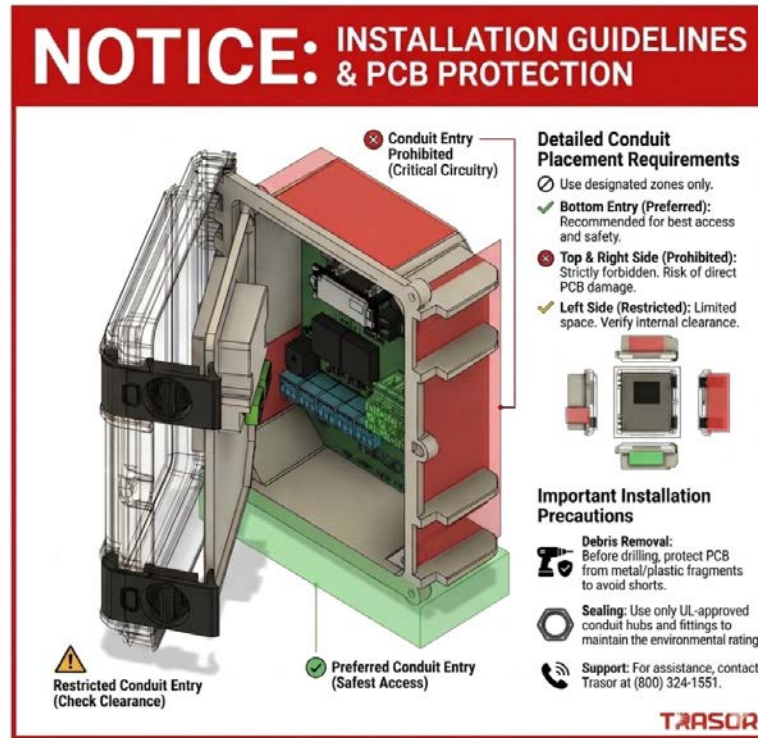
- Do not operate the unit if damaged or compromised.
- Use only Trasor - approved components and accessories.
- Observe proper ESD precautions when handling internal electronics.

1.6 Improper Use

- Use this product only as specified by Trasor Corp.
- Unauthorized modifications or improper installation may:
 - Void warranty
 - Compromise safety
 - Damage equipment

1.7 Final Safety Notice

Failure to follow these instructions may impair the protection provided by this equipment and result in hazardous conditions.



2.1 Conduit Entry & PCB Protection

To ensure the integrity of the internal electronics and maintain the enclosure's environmental rating, installers must strictly adhere to the following conduit entry requirements. The **HT-PRO MON** unit contains a precision printed circuit board (PCB); improper drilling or entry will result in permanent damage to the device.

2.2 Conduit Placement Requirements

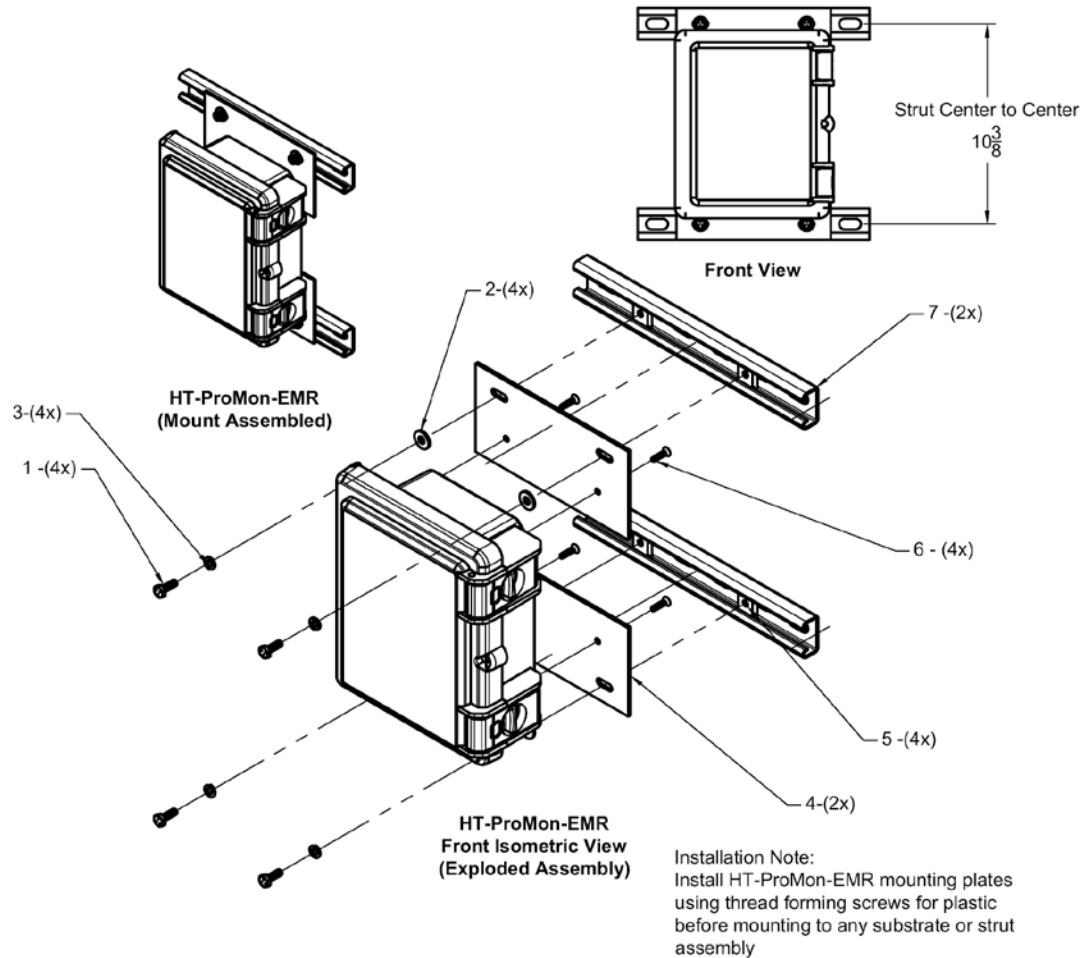
Drilling or punching holes outside of the approved zones may interfere with the PCB mounting or internal components.

- Bottom Entry (Preferred):** All conduit connections should be made from the **bottom side** of the enclosure.
- Top & Right Side (Prohibited):** Do **not** enter the enclosure from the **top** or **right** sides, as these areas are strictly prohibited and may cause direct contact with critical circuitry.
- Left Side (Restricted):** Entry from the **left side** is partially prohibited. Verify internal clearances before proceeding.

2.3 Important Installation Precautions:

1. **Debris Removal:** Before drilling, ensure the PCB is protected from plastic shavings or metal fragments, which can cause electrical shorts.
 2. **Sealing:** Use only UL-approved conduit hubs and fittings to maintain the enclosure's seal.
- **Support:** For technical assistance regarding specific entry points or clearances, contact **Trasor** at **(800) 324 -1551**

3.1 Mounting Assembly Diagram



Item Number	Description	Quantity
1	Hex Screw - Nickle Plated Steel, 1/4-20 Thread Size, 3/4" Length	4
2	1/4" Washer, Zinc Plated Steel	
3	1/4" Split Lock Washer	4
4	HT-ProMon-EMR Mounting Plate	2
5	Strut Channel Nut, Zinc-Plated Steel, 1/4"-20 Thread Size, 5/32" High	4
6	Stainless Steel Flat Head Thread-Forming Screws for Plastic, Phillips Drive, Number 10 Screw Size, 3/4" Long	4
7	Strut Channel, Low-Profile, Slotted Hole, Zinc-Plated Steel	2
8	HT-ProMon-EMR	1

3.2 Mounting Instructions

General

Install the HT-ProMon-EMR controller in accordance with all applicable national and local electrical codes, including NEC (NFPA 70).

Installation shall be performed by qualified personnel.

Mounting Plate Installation

Secure the mounting plates to the enclosure using the provided thread-forming screws.

Tighten screws evenly to ensure proper seating of the mounting plates.

Do not overtighten. Damage to the enclosure may result.

Mounting Preparation

Install strut channel nuts at the intended mounting location.

Verify that mounting hole spacing aligns with the mounting plates.

Controller Positioning

Position the controller against the mounting surface and align mounting plate holes with the installed strut nuts.

The enclosure shall be mounted such that:

- Conduit entry is oriented downward (preferred)
 - The enclosure door can be fully opened
 - Adequate clearance is provided for wiring and servicing
-

Securing the Controller

Install mounting screws through the mounting plates into the strut nuts.

Where provided, install flat washers and lock washers.

Tighten all fasteners securely.

Mechanical Verification

Verify that:

- The controller is securely fastened
 - The assembly is rigid and free from movement
 - The enclosure is not distorted
-

Final Inspection

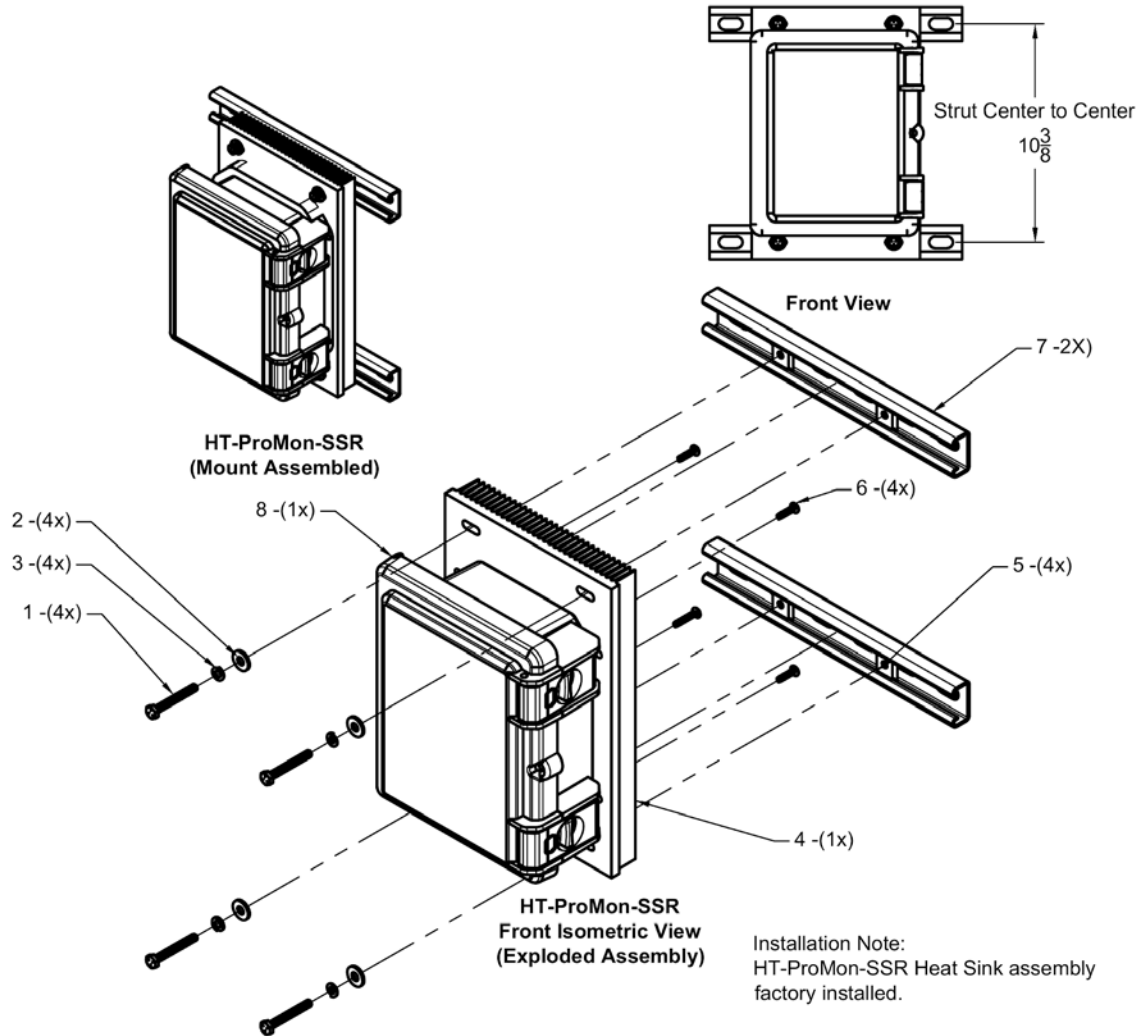
Prior to energizing the unit:

- Verify all mounting hardware is properly installed and tightened
 - Verify enclosure integrity has not been compromised
 - Verify compliance with conduit entry requirements as specified in this document
-

Installation Note

Mounting plates shall be installed to the enclosure prior to securing the unit to any mounting surface.

4.1 Mounting Assembly Diagram



Item Number	Description	Quantity
1	Hex Screw - Nickle Plated Steel, 1/4"-20 Thread Size, 2" Length	4
2	¹ / ₄ " Washer, Zinc Plated Steel	
3	¹ / ₄ " Split Lock Washer	4
4	HT-ProMon-SSR Mounting Plate	1
5	Strut Channel Nut, Zinc-Plated Steel, 1/4"-20 Thread Size, 5/32" High	4
6	Stainless Steel Flat Head Thread-Forming Screws for Plastic, Phillips Drive, Number 10 Screw Size, 3/4" Long	4
7	Strut Channel, Low-Profile, Slotted Hole, Zinc-Plated Steel	2
8	HT-ProMon-SSR (Heatsink Assembled)	1

4.2 Mounting Instructions

General

Install the HT-ProMon-SSR controller in accordance with all applicable national and local electrical codes, including NEC (NFPA 70).

Installation shall be performed by qualified personnel.

Mounting Plate Installation

Secure the mounting plate to the enclosure using the provided thread-forming screws.

Tighten screws evenly to ensure proper seating of the mounting plate.

Do not overtighten. Damage to the enclosure may result.

Mounting Preparation

Install strut channel nuts at the intended mounting location.

Verify that mounting hole spacing aligns with the mounting plate.

Controller Positioning

Position the controller against the mounting surface and align mounting holes with the installed strut nuts.

The enclosure shall be mounted such that:

- Conduit entry is oriented downward (preferred)
 - The enclosure door can be fully opened
 - Adequate clearance is provided for wiring and servicing
 - Adequate clearance is provided around the heat sink for airflow
-

Securing the Controller

Install mounting screws through the mounting plate into the strut nuts.

Where provided, install flat washers and lock washers.

Tighten all fasteners securely.

Mechanical Verification

Verify that:

- The controller is securely fastened
 - The assembly is rigid and free from movement
 - The enclosure and heat sink are not distorted or obstructed
-

4.3 Thermal Considerations

The HT-ProMon-SSR includes an integral heat sink.

The unit shall be installed such that:

- Heat sink fins are not obstructed
- Adequate airflow is maintained around the heat sink
- The unit is not installed in a manner that restricts heat dissipation

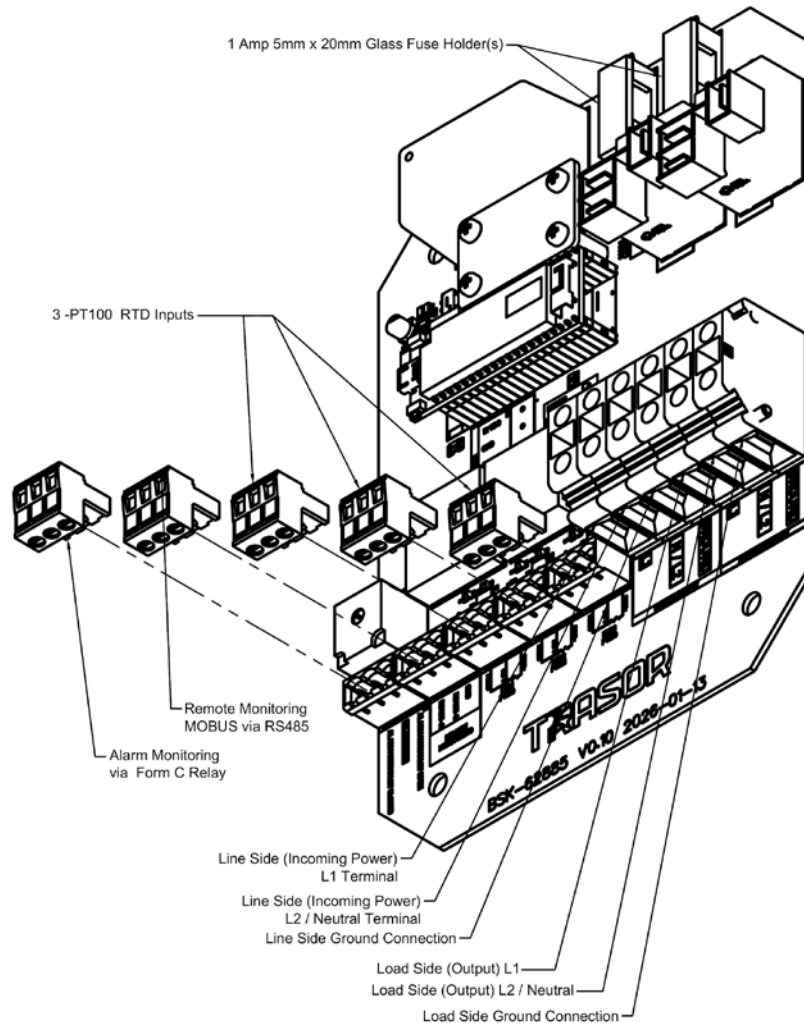
Failure to provide adequate airflow may result in overheating and reduced product life.

Final Inspection

Prior to energizing the unit:

- Verify all mounting hardware is properly installed and tightened
 - Verify enclosure integrity has not been compromised
 - Verify compliance with conduit entry requirements as specified in this document
-

5. Terminal Block Description



The HT-ProMon Series utilizes a combination of fixed and removable terminal blocks to facilitate wiring connections. The low -voltage input connections —including the RTD sensor inputs, RS-485 remote monitoring port, and alarm relay —are made using removable (plug-in) terminal blocks. These connectors allow the wiring harness to be disconnected from the control board without disturbing individual wire terminations, simplifying field replacement and servicing. Each removable connector is keyed to prevent in correct reinsertion. When reconnecting, ensure the connector is fully seated and latched. The supply (line side) and load side terminals are fixed screw-type terminals rated for 14 –8 AWG copper conductors.

6. Wiring Instructions

DANGER — DISCONNECT AND LOCK OUT ALL POWER BEFORE MAKING ANY WIRING CONNECTIONS.

All wiring connections are made at the terminal blocks located on the printed circuit board backplane.

Open the enclosure cover and rotate the swing pan outward to access the terminal blocks.

Refer to Figure 5 for terminal block locations.

6.1 General Requirements

Installation shall be performed by qualified personnel in accordance with all applicable national and local electrical codes, including NEC (NFPA 70).

Copper conductors only shall be used.

Conductors shall be sized in accordance with applicable codes and terminal ratings.

6.2 Terminal Block Reference

Terminal Block	Terminals	Wire Range	Description
Line Side Input	L1, L2/N, G	14–8 AWG	Single-phase AC supply (85–305 VAC)
Load Side Output	<i>(See enclosure label)</i>	14–8 AWG	Switched output to heat trace load
Alarm Relay	NO, COM, NC	28–16 AWG	SPDT relay, 6A @ 250VAC / 24VDC max
Remote Monitoring Port	RS485A, RS485B, GND	28–16 AWG	2-wire RS-485 Modbus RTU
RTD 1	Force+, RTD+,	28–16	3-wire PT100 input

	RTD–	AWG	
RTD 2	Force+, RTD+, RTD–	28–16 AWG	3-wire PT100 input
RTD 3	Force+, RTD+, RTD–	28–16 AWG	3-wire PT100 input

6.3 Supply & Load Wiring

Connect incoming power to the **Line Side Input** terminals.

Connect the heat trace circuit to the **Load Side Output** terminals.

The controller switches both supply conductors (**L1 and L2/N**) to the load.

The equipment grounding conductor shall be connected to the **G (ground)** terminal.

Verify that:

- Supply voltage is within the specified range
- All conductors are properly terminated
- Branch circuit protection is provided in accordance with applicable codes

6.4 Supply Configuration

The controller switches both supply conductors to the load.

- For 120 VAC systems, the controller switches:
 - L1 (Hot) and L2/N (Neutral)
- For 208/240 VAC systems, the controller switches:
 - L1 (Line) and L2 (Line)

For 208 VAC and 240 VAC applications, both supply conductors (L1 and L2) are energized line conductors.

Both conductors shall be connected to the Line Side Input and switched through the controller.

Do not connect a neutral conductor in place of L2 for 208/240 VAC systems.

6.5 Load Current & Output Types

The HT-ProMon controller is available in Electromechanical Relay (EMR) and SolidState Relay (SSR) configurations.

EMR Models

EMR outputs are rated for a maximum load current of **30 amperes per output circuit** .

The load current shall not exceed this rating.

SSR Models

SSR outputs utilizesolid -state switching with time -proportioning (duty cycle) control to regulate power delivered to the heat trace load.

The maximum allowable load current for SSR models shall be as specified on the product nameplate.

6.6 SSR Thermal Requirements

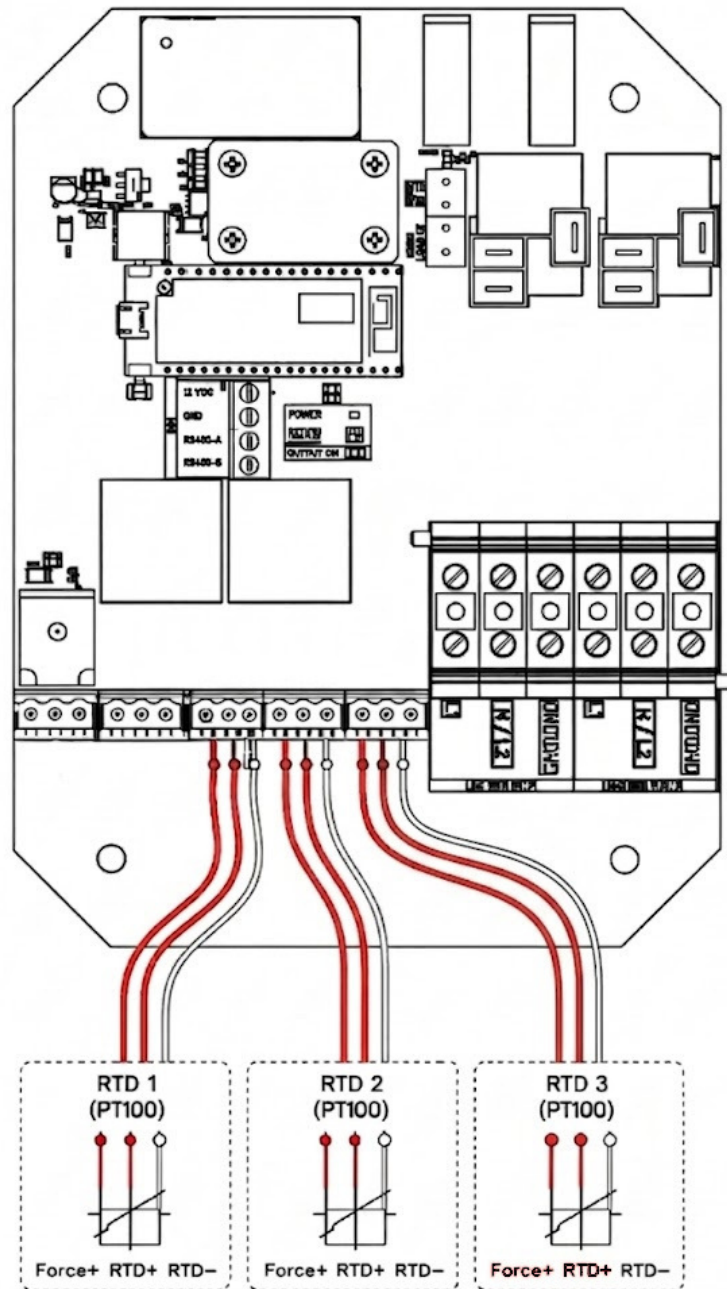
SSR models generate heat during operation.

The unit shall be installed such that:

- Adequate airflow is provided around the heat sink
- Heat sink surfaces are not obstructed
- Installation does not restrict heat dissipation

Failure to provide proper thermal management may result in overheating or equipment damage.

7.1 RTD Wiring Diagram



7. RTD Sensor Wiring

A minimum of one RTD (**RTD 1**) shall be connected and configured as the control sensor.

At least one RTD input shall be active for proper controller operation.

7.2 3-Wire PT100 Connection

Terminal	Connection
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Force+	One of the two identical (same - color) leads
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RTD+	The second identical lead
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RTD-	The third (different - color) lead
------	------------------------------------

Ensure:

- Proper lead identification is maintained
 - All connections are secure
 - Shielding (if used) is grounded at one end only
-

8.1 RS 485 Remote Monitoring

Connect RS-485 communication wiring to the **Remote Monitoring Port** .

- RS485A → Non-inverting (+)
- RS485B → Inverting (-)
- GND → Signal reference (optional but recommended)

Use twisted pair cable suitable for RS-485 communication.

8.2 Alarm Relay Wiring

The alarm relay provides a **Form C (SPDT) dry contact output** .

- **NO** — Normally Open
- **COM** — Common
- **NC** — Normally Closed

Connected circuits shall not exceed:

- 6A @ 250 VAC / 6A @ 24 VDC
-

8.3 Internal Fuses

The control power supply is protected by **two (2) replaceable 5 mm × 20 mm, 1A fuses** located on the control board.

If the controller does not power on:

1. De-energize and lock out all power
 2. Open enclosure and rotate swing pan
 3. Inspect fuses for continuity
 4. Replace with identical type and rating only
-

8.4 Final Wiring Verification

Prior to energizing the unit:

- Verify all wiring connections are correct and secure
- Verify proper grounding has been established
- Verify no loose conductors or debris are present
- Ensure swing pan is returned to the closed position
- Ensure enclosure cover is properly secured